Reg. No. | | | | | | | | | | |

[16CS661]

SIXTH SEMESTER B. TECH EXAMINATION, JUNE-2019

SYSTEM SIMULATION AND MODELING

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full questions choosing atleast two questions from each part.
2. Missing data may be suitably assumed.
3. Random Digits/Normal numbers tables may be provided.

PART-A

1. a. What is Simulation? Explain with flowchart, the steps involved in simulation study. **09**

   b. When simulation is an appropriate tool? And when is not? **05**

2. a. Explain the concept of world view with its different types. **07**

   b. Explain simulation with a neat block diagram for the single server queue simulation. **07**

3. a. Hurricane hitting east coast India follows Poisson with a mean of 0.8 per year. Determine **09**

   i. The probability of more than 2 hurricanes in one year.

   ii. The probability of exactly one hurricane in one year.

   iii. The probability of hurricane not hitting in a year.

   b. A certain program component has a triangular distribution with a=0.15, b=2.5 and c=4.5. Find out probability for a random program component is 3.5 and less. **05**

4. a. Explain the various steady state parameters of M/M/1. **05**

   b. Explain the characteristics of a queuing system. **06**

   c. Customers arrive at random to a vehicle service shop at a rate of λ=40 customers per hour. Assume that there are 10 clerks, each serving μ=5 customers per hour on the average. In this long run, find average number of busy servers. **03**

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PART-B

5. a. What considerations are to be taken care in generating random numbers? **05**

b. Generate three Poisson variates with mean $\alpha=0.25$. The random numbers are 0.073, 0.693,0.945,0.739,0.014,0.342 **09**

6. a. List out the steps involved in the development of model of input data. **04**

b. Records pertaining to the monthly number of job related injuries at an underground coal mine were being studied by a federal agency. The values for the past 100 months were as follows. **10**

Injuries/month	Frequency of occurrences
0	35
1	40
2	13
3	6
4	4
5	1
6	1

i. Apply chi-square test to these data to test the hypothesis that the underlying distribution is Poisson. Use $\alpha=0.05$

ii. Apply chi-square test for Poisson distribution with mean=1.0 and 0.05

7. a. Explain the types of simulation with respect to output analysis. **06**

b. Explain output analysis for terminating simulation. **08**

8. a. Explain with a neat diagram, model building, verification and validation process. **08**

b. Write short notes on **06**

i. Optimization via simulation

ii. CPU simulation